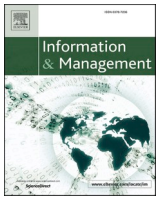
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The marketing effects of live streaming in online marketplaces

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ABSTRACT

With the development of live-streaming e-commerce, brands are experimenting with live streaming as a new sales channel. However, the real-time nature of live streaming, coupled with promotional content and a large volume of products, may overwhelm viewers and diminish the brand's benefits. From the dual perspectives of information overload theory and trust theory, this paper utilizes the sales data of beauty and skincare products during the "Double 11" shopping festival of Taobao in the year 2020 and examines the effects of the duration of product introduction and the number of listed products on sales performance for different live-streaming modes. The empirical results reveal that the duration of product introduction affects brand online stores' sales performance positively, while the number of listed products affects sales performance negatively. In addition, compared with Internet celebrity live streaming, the positive effect of the duration of product introduction on sales performance is stronger in the brand shop live-streaming mode. Compared with brand shop live streaming, the negative effect of the number of listed products on sales performance is stronger in the Internet celebrity live-streaming mode.

1. Introduction

With the development of live-streaming technology and the widespread use of mobile devices, live-streaming e-commerce has become a new business trend [1,2]. According to an industry report [3], the market size of the live-streaming e-commerce market in China reached 5.3 trillion yuan in 2024 and it is expected to exceed 8.2 trillion yuan in 2026. As an innovative type of e-commerce, live-streaming e-commerce offers real-time visibility and synchronous interactivity [4]. Streamers can show products from different perspectives, answer customer questions in real time, and provide personalized guidance and services [5]. This greatly reduces the uncertainty perceived by consumers, stimulating them to buy on the spot [6]. It is reported that the purchase rate of live-streaming e-commerce is 2–3 times higher than that of traditional e-commerce [7]. For example, Jiaqi Li, a popular Chinese streamer, made a record of selling 15,000 lipsticks in five minutes. Recognizing the huge benefits, many brands are experimenting with live streaming as a new sales channel.

The exponential growth of live-streaming e-commerce, as previously discussed, has created an engaging marketplace. However, this proliferation of content also brings forth the challenge of information overload. The real-time nature of these live streaming, coupled with the high volume of product and promotional content, can potentially overwhelm viewers and diminish the effectiveness of the platform. While existing research has examined various factors influencing live-streaming shopping, such as the average price of products, number of followers/fans, and the attractiveness of streamers [8–10], a crucial aspect that remains relatively underexplored is the impact of information overload. This study seeks to address this gap by examining how information overload manifests in the live-streaming context and its implications for sales performance.

Within the live-streaming e-commerce context, consumers are presented with a continuous stream of product information disseminated by streamers, profoundly influencing their information processing and purchase decisions, and raising the risk of information overload. The information overload theory indicates that the quality of consumer decision-making is influenced by the amount of information received and the number of options available [11–13]. Therefore, in this study, we focus on two major factors that may lead to information overload among viewers: the duration of product introduction and the number of listed products. For the first factor, streamers' introduction is a crucial way for viewers to learn about the products in live streaming. As detailed product information helps to reduce the perceived risk of product quality, the duration of product introduction is closely related to

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0378-7206/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

the completeness and quality of information obtained by viewers [14]. For the second factor, within the dynamic and interactive context of live commerce, a large number of products may lead to higher search costs and information overload for customers [15,16].

In addition, brand stores generally choose two live-streaming modes, Internet celebrity live streaming and brand shop live streaming (Table 1), to generate live-streaming revenue [4]. Internet celebrity live streaming is also known as KOL (key opinion leader) live streaming. Typically, the streamers are authorities with a large number of fans in their chosen product categories. For instance, some streamers excel in selecting clothing for their customers or providing beauty advice. In a live-streaming session lasting 3–4 h, Internet celebrities often promote dozens of brands by recommending each product to their fans through professional introductions and visual presentations or by trying them on themselves. As a result, such promotion activities induce conformity effects and encourage viewers to make impulse purchases [6]. In contrast to Internet celebrity live streaming, streamers hired by online stores for live streaming are generally not as well-known as celebrities and they only work as salespersons for one brand name [8,17]. In effect, brands eliminate middlemen (i.e., KOLs) in brand shop live streaming, and all live-streaming sales links (such as product portfolios, promotion information, and after-sale services) are under the direct control of the brands, which guarantees the quality of the live-streaming services [18]. By leveraging the interactivity and synchronicity of live streaming, brands can provide authoritative and personalized services to customers, which are conducive to brand promotion [4].

The distinctiveness of these two live-streaming modes gives rise to two fundamentally different types of trust. While trust in Internet celebrities is based on the perceived attractiveness, trustworthiness, and professionalism of the individual KOL (often an indirect form of trust, where trust in the person, who might be an actor, celebrity, or influential blogger, transfers to the product they endorse [6,19]), trust in Internet brand shops is rooted in consumers' perceptions of the brand's product quality, service capabilities, and overall reliability (a more direct form of trust based on the brand's attributes [20]). As mentioned above, in the brand shop live-streaming mode, the brand retains the ultimate authority to interpret policies related to product quality and after-sales service [18]. This highlights the heightened reliability associated with placing trust in brand-owned online stores, consequently indicating that the effects of trust in Internet brand shops manifest with greater potency than those of trust in Internet celebrities. When consumers exhibit a high degree of trust toward the product under consideration, the detrimental effects of information overload on their purchase decision-making process are mitigated. In other words, by relying on established trust, consumers can alleviate their cognitive processing burden and reduce informational dependence, thereby streamlining their purchase decision [21,22]. Based on this, researchers should consider the live-streaming mode and its impact on consumer decisions, assisting brand stores in achieving better live-streaming sales performance.

In summary, this paper aims at addressing the following research questions: (1) *How do the duration of product introduction and the number of listed products affect brand online stores' live-streaming sales performance?* (2) *Do these effects differ between the two live-streaming modes (Internet celebrity live streaming and brand shop live streaming)?* To this end, we utilized the sales data of beauty and skincare products during the "Double 11″ shopping festival of Taobao in the year 2020 and examined the effect of the duration of product introduction and the number of listed products on sales performance in different live-streaming modes.

This study aims to make the following contributions. First, based on the theory of information overload, we examined the effects of the duration of product introduction and the number of listed products, which are strongly correlated with consumers' information processing capability. This work enriches the extant live-streaming e-commerce literature by advancing the contextual application of information overload theory to the unique setting of live-streaming commerce, moving beyond mere theoretical debates to targeted empirical exploration. Second, acknowledging that the literature on trust has proposed various typologies [21,63], we contend that the unique dynamics of the live-streaming e-commerce context calls for a new framework. Therefore, we systematically distinguish and unpack the distinct trust mechanisms operating in the two primary live-streaming modes. This extends the theory of online trust from "search" and "browsing" based e-commerce to the more interactive and synchronous selling format of live-streaming e-commerce. Third, and most critically, this paper integrates information overload theory and trust theory to construct a holistic predictive framework. We posit that the live-streaming mode activates a specific trust mechanism, which critically mediates the effects of information overload on consumer outcomes. This integrative lens clarifies the boundary conditions for information overload—explaining when it harms consumer decision-making and when the harm is mitigated. Consequently, this study provides a more nuanced theoretical understanding of consumer behavior in live streaming, offering valuable, context-specific insights for both academic research and managerial practice.

2. Theoretical background

2.1. Live-streaming e-commerce

Most existing research on live-streaming e-commerce has employed the survey method to explore the antecedents of consumers' purchase intentions, including factors related to information technology (e.g., visibility and interactivity), consumers (e.g., perceived values), products (e.g., product quality), and streamers (e.g., attractiveness and professionalism, streamer and product matching; See Table 2). Sun et al. [23] built a theoretical model from the perspective of IT affordance and found that the visibility, meta-voicing, and guidance shopping affordance of live streaming can influence customers' purchase intentions through interaction. Wongkitrungrueng and Assaraut [6] examined the impacts of symbolic value, utilitarian value, and hedonic value on customer engagement from the perspective of value and trust. Cai et al. [24] found

Table 1
Internet celebrity live streaming vs. brand shop live streaming.

	Internet Celebrity Live Streaming	Brand Shop Live Streaming
Ownership	Internet celebrity	Brand merchants
Typical representatives	Jiaqi Li (on Taobao), Yonghao Luo (on TikTok), Ba Xin (on Kwai)	Adidas, Teenie Weenie, Estee Lauder (the monthly live-streaming sales revenue of these brands have exceeded 100 million RMB in China)
Streamers	Internet celebrities, movie stars, KOLs	Brand employees or full-time streamers hired through third-party operating companies
Products	Products of many brand names with limited display time for individual products	Products of a single brand name with detailed product introduction
Advantages	The fame and charm of streamers make it easier to attract consumers to watch and participate in live streaming	It provides customers with higher-quality products and professional services; also, the company may have lower marketing costs
Challenges	Some viewers watch purely out of curiosity and lack the motivation to make a purchase	Low exposure and limited traffic

Table 2
Literature review on live-streaming e-commerce.

Previous Research	Antecedents	Outcomes	Theory	Research Methods	Two Live-Streaming Modes	Information Overload
Sun et al. [23]	IT affordance (visibility, meta-voicing, guidance shopping), immersion, presence	Purchase intention	Affordance theory	Survey	No	No
Wongkitrungrueng and Assarut [6]	Symbolic value, utilitarian value, hedonic value, and trust	Consumer engagement	Perceived value, trust theory	Survey	No	No
Cai et al. [24]	Utilitarian motivations, hedonic motivations	Watching intention	Motivation theory	Survey	No	No
Chen et al. [25]	Product quality uncertainty, product fit uncertainty, habit	Purchase intention	Dual-process theory	Survey	No	No
Park and Lin [26]	Internet celebrity–product fit, live content–product fit, source attractiveness, source trustworthiness	Purchase intention	Source credibility theory and match-up hypothesis	Survey	No	No
Meng et al. [27]	Social presence, information source characteristics of Internet celebrity	Purchase intention	Psychological arousal theory, source credibility theory	Survey, interview, secondary data analysis	No	No
Huang et al. [28]	Streamer type	Purchase intention	Social influence theory	Laboratory experiment, secondary data analysis	Yes	No
Liu and Yu [29]	Streamer type, product type, and brand awareness	Purchase intention	SOR model	Laboratory experiment	Yes	No
Wang et al. [8]	Streamer type	Intention to follow a live-streaming brand community	Motivational theory	Laboratory experiment, secondary data analysis	Yes	No
Zhang et al. [30]	The total number of products introduced and the audience's average stay time in a live stream	Live-streaming sales	Information overload	Secondary data analysis	No	Yes
This Paper	Duration of product introduction, number of listed products	Live-streaming sales	Information overload, trust theory	Secondary data analysis	Yes	Yes

that utilitarian and hedonic motivations are all related positively to customers' watching and purchase intentions. Based on the dual-process theory, Chen et al. [25] proposed a model based on a signal mechanism and an automatic habit mechanism and found that product quality and behavioral habits have a significant impact on consumer purchase intention. Grounded on source credibility theory and the match-up hypothesis, Park and Lin [26] found that source trustworthiness, hedonic attitude, and self–product fit contribute to purchasing intentions. Meng et al. [27] argued that for Internet celebrity streamers, the most significant characteristics that affect consumers are trustworthiness, professionalism, and interactivity. Although these studies have deepened our understanding of live-streaming e-commerce, they have overlooked the impact of the information characteristics of live streaming on sales performance, especially the duration of product introduction and the number of listed products, which are strongly correlated with consumers' information processing capability in the context of live streaming. Similarly, Zhang et al. [30] explored the effects of the number of products and the viewer's average stay time on live-streaming sales. However, they neglected to consider the distinct characteristics of different live-streaming modes, which presents an opportunity for us to explore the effects of information overload across different contexts.

Furthermore, the popularity of brand shop live streaming in e-commerce has drawn the attention of the academic community. Wang et al. [31] claimed that in brand shop live streaming, customers and sellers are spatially separated yet temporally synchronous. This enables customers to seek immediate purchase recommendations directly from firms. Some studies have compared Internet celebrity live streaming and brand shop live streaming in terms of the characteristics of streamers. According to Huang et al. [28], brand shop streamers possess professional knowledge of products, which can enhance consumers' perception of product quality. They are more likely to arouse consumers' purchasing intentions for utilitarian products than Internet celebrity streamers. Internet celebrity streamers are better positioned in promoting hedonic products, since they may appeal to consumers by leveraging their own popularity. Liu and Yu [29] found that influential streamers and well-known brands are more diagnostic to consumers and lead to stronger purchase intentions. From the perspective of value appeal, Wang et al. [8] argued that the professionalism of brand shop streamers provides informational value, which makes consumers more likely to follow a livestream brand community, especially for utilitarian products. The attraction of celebrity streamers provides emotional value, which makes consumers more likely to follow a livestream brand community, especially for hedonic products. It is worth noting that these studies emphasize the factors of streamers in the two live-streaming modes but overlook the impact of different consumer trust mechanisms, stemming from the distinctions between these modes, on sales outcomes.

In summary, this paper considers two information characteristics in live streaming, the *duration of product introduction* and the *number of listed products*, which have barely been examined in the literature. Furthermore, it examines how the duration of product introduction and the number of listed products affect sales performance, and how these effects vary between different live-streaming modes. Finally, to address the issue of inaccurate data caused by recall bias in questionnaire and survey methods, it explores the marketing effect using real-world data, including the duration of product introduction, the number of listed products, and live-streaming sales.

2.2. Information overload

Information overload is a topic relevant to a wide range of disciplines. It is based on the argument that every individual has limited cognitive capability, so when the amount of information exceeds the recipient's cognitive capacity, information overload occurs [32]. The phenomenon of information overload is known by various names, including information overabundance [33], information fatigue [34], social media overload [35], communication overload [36], and

cognitive overload [37]. According to Klausegger et al. [38], there is an inverse U-shaped relationship between decision outcomes and information load. This suggests that both insufficient information and information overload can reduce decision quality, whereas a moderate amount of information can lead to better outcomes [32,38].

In the field of consumer behavior, information overload has been studied extensively. Specifically, information overload in consumer decision-making can be caused by the number of alternatives or the number of product attributes [11–13]. Many studies on decision-making processes have examined choice overload. Scheibehenne et al. [39] found that when consumers are faced with many options, they can only choose one in the end. The more choices consumers have to forgo, the higher the opportunity cost, which leads to a decrease in satisfaction with the purchased goods. Existing research also highlights the role of the amount of information. Lurie [40] asserts that when consumers are confronted with a large amount of product information, they want to utilize information resources to aid in their judgments and choices. However, when the amount of product information exceeds their cognitive capacity, consumers may experience negative emotions.

We anticipate that consumers in live streaming will experience a similar phenomenon. The streamer's introduction of the product is a vital source of information for viewers. The duration of product introduction relates directly to the amount of product information that consumers can acquire. The number of listed products refers to the total number of products introduced by streamers in a live-streaming show. In other words, the number of listed products is the total number of product choices available to consumers during a single live-streaming session. Therefore, based on the theory of information overload, we will examine the impact of the duration of product introduction and the number of listed products on brand stores' sales performance.

2.3. Trust

Trust is the belief that a relevant party will fulfill its obligations in an appropriate and ethical manner, thereby meeting the expectations of the trusting party [41]. Within the process of e-commerce transactions, trust can mitigate the risks and uncertainties arising from information asymmetry, thereby facilitating the successful completion of the transactions [21,22]. Previous research has shown that online trust has a significant impact on consumer behavioral intentions, even in the context of live streaming [42–44]. Chen et al. [45] found that trust influences viewers' reward behavior positively, such as giving the streamers money or "likes." Xu et al. [22] indicated that trust plays an intermediary role between the professionalism of streamers and the viewers' willingness to purchase. However, the trust mechanisms inherent in the two live-streaming modes may differ, as explained below.

2.3.1. Trust in an internet celebrity

Internet celebrities, in their role as temporary endorsers, strengthen viewers' positive feelings, which in turn leads to the development of trust in products [6]. The theory of trust posits that trust in one entity can be transferred to an associated, yet unknown, person or entity [46,47]. This implies that once consumers trust the celebrity streamer, their positive sentiments and evaluations toward the streamer are transferred to the brand and product through psychological linkages, ultimately fostering trust in the product [19]. Park and Lin [26] also demonstrate that consumers' positive attitudes and trust toward a streamer are transferred to the products they endorse. That is, if consumers genuinely trust a live streamer, they are more likely to develop trust in the products promoted by the streamer, thereby enhancing their purchase intentions.

2.3.2. Trust in a brand shop

Brand trust is a critical factor in consumers' decisions to buy, use, or engage with products and services. It is defined as a consumer's perceived security in a brand's ability to fulfill its promises [20,48]. This perception encompasses the belief that the brand is reliable, capable of addressing customer needs, and unlikely to engage in opportunistic behavior by exploiting customer vulnerabilities [49]. That is, consumers believe that the brand's products will meet expectations and that their appearance and functions align with the brand's claims. Existing research has demonstrated that brand trust can enhance consumer purchase intentions [50,51]. In the brand shop live-streaming mode, brands can engage directly with consumers, enabling them to build closer relationships more rapidly, strengthen consumer trust in the brands and their products, and consequently achieve better sales performance.

In addition, our analysis differentiates trust in an Internet celebrity from trust in a brand along key dimensions such as the object of trust and the foundation of trust. The details are presented in Table 3.

3. Research model and hypotheses development

This study focuses on the impact of the duration of product introduction and the number of listed products on brand stores' sales performance. In addition, we tested the moderation effects of the live-streaming mode empirically. The research model is shown in Fig. 1.

3.1. The main effects

3.1.1. The duration of product introduction

The duration of product introduction refers to the total time spent in promoting a product in the live-streaming room, including the time in introducing the brands, product details, marketing campaigns, and return policies of a particular product. Due to the virtual nature of live-streaming e-commerce, the streamer's introduction of the product is a crucial way for viewers to obtain information about products. The more detailed and higher-quality the product information provided by the streamer, the more persuasive it is perceived by the customers. This leads to a higher evaluation of the product and an increased willingness to buy [52]. Therefore, in the process of product recommendation, streamers should strive to provide detailed information on the brand and products, including the brand story, product features, past consumers' usage experience, discounts and/or promotional information, and

Table 3
Trust in the Internet celebrity vs. trust in the brand.

	Trust in the Internet Celebrity	Trust in the Brand
Trust object	Individual internet celebrity (e.g., live-streamers, KOLs, influencers)	Brand entity (e.g., enterprises, brand IPs)
Foundation of Trust	Based on "emotional connection + content resonance," focusing on "personalized attraction"	Based on "long-term value + stable commitment," focusing on "functional/value-based assurance"
Formation path	Social media interaction, personal storytelling	Long-term market performance, product quality, customer service, and branding
Establishment cycle	Short and efficient (trust can be accumulated via viral content and frequent interaction within weeks or months)	Long-cycle (trust demands long-term, consistent effort in product quality, user feedback, and crisis management)
Consumer loyalty	Tied to the individual, it may fade with influence	Enduring, based on consistent brand experience
Risk sensitivity	Extremely high (personal risks such as persona collapse and false advertising directly shatter trust)	Relatively low (isolated issues, like product defects, can be resolved by PR and after-sales support, preserving overall trust)
Example	Followers buy products promoted by a celebrity influencer	Consumers choose a brand due to its proven quality and reputation

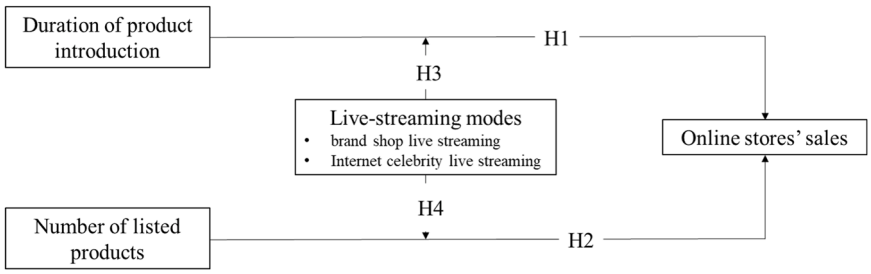


Fig. 1. The research model.

delivery services. Furthermore, in a traditional e-commerce model, excessive product information can easily lead to information overload, since it requires processing capability [53]. However, in the live-streaming e-commerce model, the streamer's product introduction helps consumers manage product information easily, reducing their overload levels and influencing their purchasing decisions positively [54]. Based on the analyses above, the following hypothesis is proposed:

H1. The duration of product introduction affects brand online stores' sales performance positively.

3.1.2. The number of listed products

The number of listed products refers to the total number of products introduced by streamers in a live-streaming show. This is also the number of available options for consumers in a live shopping scenario. The real-time interactive environment of live-streaming commerce induces a state of high cognitive arousal, which favors impulsive purchases over systematic information processing [4,6]. While this environment is conducive to quick decisions, it also makes viewers' cognitive resources exceptionally scarce and valuable. Unlike traditional e-commerce, where consumers can deliberate at their own pace, the fast-paced and ephemeral nature of live streaming places a premium on the viewer's limited attention and processing capacity.

In this context of cognitive scarcity, an extensive product inventory is not an asset but a significant liability. Each additional product presented competes for the viewer's already strained attention, contributing directly to information overload and accelerating the onset of decision fatigue [30]. Faced with a barrage of options and insufficient time to evaluate them systematically, consumers are less likely to make a confident purchase decision [55–57]. Instead of stimulating desire, an overabundance of choices can lead to choice deferral—where viewers postpone their decisions to a later time—or even choice avoidance, causing them to exit the live stream altogether [10]. This phenomenon, known as the "paradox of choice," is particularly amplified in the high-pressure, time-constrained setting of live streaming [58].

Therefore, in the live-streaming commerce context, the number of listed products is expected to have a direct and negative impact on sales performance. As the product count increases, the cognitive burden on viewers intensifies, leading to diminished purchase conversion rates and lower overall sales. Based on the analyses above, the following hypothesis is proposed:

H2. The number of listed products affects brand online stores' sales performance negatively.

3.2. The moderation effects of live-streaming modes

In practice, brand stores typically adopt two live-streaming modes, Internet celebrity live streaming and brand shop live streaming [4]. Internet celebrity live streaming involves influencers leveraging their personal influence, fan bases, and traffic advantages to drive product sales. Brand shop live streaming, however, is self-operated by a brand, where streamers engage in direct interaction with consumers through professional explanations [8,17]. Moreover, the inherent differences between the two distinct live-streaming modes imply that the underlying mechanisms or antecedents through which viewers develop trust may differ. Within the Internet celebrity live-streaming mode, the celebrity streamer assumes the role of a temporary endorser, acting as an intermediary between the product and the viewer [6]. In essence, this mode operates through a mechanism of trust transfer. Here, consumers infer the quality of the product by placing their trust in the individual serving as the streamer (such as an actor, celebrity, or well-known blogger) [19,26]. However, the brand shop live-streaming mode operates on direct trust. This process mirrors the development of brand trust in traditional business modes, where consumers place direct trust in a brand regarding the quality of its products and the reliability of its services [20].

Differences in the trust mechanisms employed by these two modes could account for the differing effects they have on information overload. Within the Internet celebrity live-streaming mode, the streamer excels at product selection as well as communication and negotiation with the brand [4,6]. Nevertheless, when product-related issues emerge, the ultimate responsibility for resolution invariably rests with the brand. This arrangement highlights the characteristics of indirect trust. In contrast, trust in brand shop live streaming derives from the reputation of brand names [20]. Furthermore, the brand assumes ultimate responsibility for both product quality and post-purchase service [18]. This direct responsibility contributes to more direct and dependable trust. Based on this, it can be concluded that trust in a brand shop exerts a stronger effect than trust in an Internet celebrity.

The duration of product introduction is crucial for consumers to acquire sufficient information, mitigating purchase risks and uncertainties [52]. However, the effectiveness of this factor varies significantly with the level of trust. In instances involving high-trust brands, an extended duration of introduction substantially magnifies favorable outcomes, resulting in a marked enhancement of sales performance. Conversely, in the context of Internet celebrity live streaming with lower trust, even prolonged product introductions struggle to alleviate consumers' doubts regarding the brand or product quality completely. Consequently, the "amplification" effect of introduction duration is considerably diminished, and its positive impact on sales becomes less pronounced. Based on the analyses above, the following hypothesis is proposed:

H3. Compared with Internet celebrity live streaming, the positive effect of the duration of product introduction on sales performance is stronger in the brand shop live-streaming mode.

As previously stated, trust in a brand shop exerts a stronger effect than trust in an Internet celebrity. When consumers hold high levels of trust in a product they are considering, the adverse effects of information overload on their purchase decision are mitigated. This demonstrates a decision heuristic effect where trust in a brand allows consumers to bypass extensive information processing, facilitating rapid purchase choices [21,22]. Hence, the following hypothesis is proposed:

H4. Compared with brand shop live streaming, the negative effect of the number of listed products on sales performance is stronger in the

Internet celebrity live-streaming mode.

4. Methodology

4.1. Data

The data were sourced from a third-party platform, Feigua Data, which aggregates and hosts live-streaming data from Taobao. We selected the beauty and skincare category, which is the primary product category of Taobao's live-streaming platform, to validate our hypotheses. Data released by Ju Meli, a social media company specializing in beauty and skincare, indicate that the sales of Taobao's beauty and skincare category achieved an annual peak in November 2020, composing approximately 25 % of the total annual sales [59]. This suggests that consumers are price-sensitive with regard to beauty and skincare products, often engaging in stockpiling during the "Double 11″ shopping festival. Consequently, the sales performance of brands during the "Double 11″ shopping festival predominantly influences their market share throughout the year. Hence, this study randomly selected 377 beauty and skincare brands that conducted live-streaming campaigns during the "Double 11" shopping festival on Taobao in 2020 to examine the marketing effects of live streaming. Within this sample of 377 brands, 138 engaged in brand shop live streaming, 227 opted for marketing campaigns within Internet celebrity live-streaming rooms, and the remaining 12 brands utilized both live-streaming modes concurrently. Considering the relatively small proportion of brands employing both live-streaming modes simultaneously, the final sample comprised 365 brands that utilized only one of the two live-streaming modes.

4.2. Variables

Dependent variables: The live-streaming sales volume (*Sales*) refers to the total sales volume of all live-streaming sessions for a specific brand during the "Double 11″ shopping festival of Taobao. If an online store chose brand shop live-streaming mode, this variable is the total sales volume of all live-streaming sessions in the brand's official live-streaming rooms. Alternatively, if the online store chose Internet celebrity live-streaming mode, this variable is the total sales volume of the brand's products in all the Internet celebrity live-streaming rooms it worked with.

Independent variables: These include the duration of product introduction (*Duration*) and the number of listed products (*Products*).*Duration* is measured by the average time spent in introducing all the products of a brand. *Products*refers to the total number of products introduced in the live-streaming room.

Moderating variable: *LiveMode* refers to the live-streaming mode chosen by the online store. When the online store chose brand shop live-streaming mode, the value of this variable is "1″; otherwise, it is "0".

Control variables: The control variables include the product price (*Price*), the number of viewers of the live streaming (*Viewers*), the number of likes of the live streaming (*Likes*), the number of fans of the live streaming (*Livefans*), the total number of products of a brand sold across all live-streaming sessions (*Brandproducts*), and the number of live-streaming sessions of the streamer or brand shop on the Taobao platform (*Livesessions*) as of the "Double 11″ shopping festival.

The description of each variable is shown in Table 4.

Table 4
Definition of variables.

	Variables	Description
Dependent variable	*Sales*	The total sales volume of all live-streaming sessions for a certain brand during the "Double 11″ shopping festival of Taobao.
Independent variables	*Duration*	The average time spent in introducing each product.
	Products	The average number of products in each live-streaming session during the "Double 11″ shopping festival.
Moderating variable	*LiveMode*	The live-streaming mode chosen by the brand shop. When the brand shop chooses the brand shop live-streaming mode, the value of this variable is "1″; otherwise it is "0."
Control variables	*Price*	The average price of all products introduced in live-streaming rooms of a focal brand.
	Viewers	The average number of viewers in each live-streaming room during the "Double 11″ shopping festival.
	Likes	The average number of likes in each live-streaming room during the "Double 11″ shopping festival.
	Livefans	The average number of fans in each live-streaming room during the "Double 11″ shopping festival.
	Brandproducts	The total number of products sold by a brand in all live-streaming rooms during the "Double 11″ shopping festival.
	Livesessions	As of the "Double 11″ shopping festival, the number of live-streaming sessions of the streamer or brand shop on the Taobao platform.

4.3. Model

To test the hypotheses posited, we utilized the ordinary least squares (OLS) regression method, which is delineated as

$$Ln(Sales) = \alpha_0 + \alpha_1 Ln(Duration) + \alpha_2 Ln(Products) + \boldsymbol{\alpha_3 CV} + \varepsilon_1, \quad (1)$$

where $Ln(Sales)$ is the natural log-transformed dependent variable, representing the logarithmic value of sales volume. $Ln(Duration)$ and $Ln(Products)$ are the natural log-transformed independent variables.$\boldsymbol{CV}$ represents a vector composed of all control variables [8–10], including *Price*, *Viewers*, *Likes*,*Livefans*,*Brandproducts*, and *Livesessions*.

In accordance with this model, the interaction terms $Ln(Duration) \times LiveMode$ and $Ln(Products) \times LiveMode$ are introduced to test the moderating effects of the live-streaming mode chosen by the brand stores. The regression equation is

$$\begin{aligned} Ln(Sales) = {} & \beta_0 + \beta_1 Ln(Duration) + \beta_2 Ln(Products) + \boldsymbol{\beta_3 CV} + \beta_4 Ln(Duration) \times LiveMode \\ & + \beta_5 Ln(Products) \times LiveMode + \varepsilon_2 \end{aligned} \quad (2)$$

5. Empirical results

5.1. Descriptive statistics and correlation analysis

Table 5 presents the descriptive statistics and correlation coefficients for the variables. The mean value of *Sales*is 20,699, and the standard deviation is 56,783, indicating that there are significant differences in live-streaming performance across beauty and skincare brands during the "Double 11″ shopping festival. The correlation analysis reveals that

Table 5
Descriptive statistics and correlation coefficients of variables.

Variable	Mean value	Standard deviation	1	2	3	4	5	6	7	8	9	10	11	12	13
1. *Sales*	20,699	56,783	1.000												
2. *LiveMode*	0.378	0.486	0.395*	1.000											
3. *Duration*	672.6	643.9	0.224*	0.697*	1.000										
4. *Products*	74.69	47.03	–0.205*	–0.442*	–0.533*	1.000									
5. *Price*	264.9	364.7	–0.022	0.081	0.172*	–0.216*	1.000								
6. *Viewers*	$2.142 \times 10^{+06}$	$6.770 \times 10^{+06}$	–0.005	–0.231*	–0.178*	0.009	0.274*	1.000							
7. *Likes*	$1.190 \times 10^{+06}$	$3.192 \times 10^{+06}$	–0.018	–0.257*	–0.188*	–0.001	0.235*	0.943*	1.000						
8. *Livefans*	$4.420 \times 10^{+06}$	$6.63 \times 10^{+06}$	0.219*	0.019	–0.038	–0.042	0.321*	0.882*	0.832*	1.000					
9. *Brandproducts*	11.98	20.71	0.637*	0.628*	0.260*	–0.268*	0.043	–0.150*	–0.171*	0.060	1.000				
10. *Livesessions*	649.5	344.6	0.259*	0.170*	–0.034	–0.124	–0.033	–0.060	–0.134	0.075	0.277*	1.000			
11. *BrandSize*	0.307	0.462	0.396*	0.706*	0.456*	–0.339*	0.193*	–0.044	–0.076	0.209*	0.534*	0.234*	1.000		
12. *Years*	12.192	8.246	0.190*	0.364*	0.322*	–0.299*	0.152*	0.099	0.094	0.180*	0.283*	–0.000	0.345*	1.000	
13. *Capital*	20,390	57,708.3	0.129	0.240*	0.150*	–0.114	–0.043	–0.028	–0.026	0.051	0.160*	–0.012	0.139*	0.458*	1.000

Notes: * $p < 0.01$, $N = 365$.

Table 6
Regression analysis.

Variables	Model 1	Model 2	Model 3
Ln(Price)	–0.350***	–0.521***	–0.550***
	(–4.74)	(–7.21)	(–7.53)
Ln(Viewers)	0.149**	0.210***	0.242***
	(2.39)	(3.48)	(3.72)
Ln(Likes)	0.160***	0.192***	0.215***
	(2.60)	(3.34)	(3.73)
Ln(Livefans)	0.632***	0.657***	0.566***
	(7.34)	(8.18)	(6.15)
Ln(Brandproducts)	1.818***	1.673***	1.611***
	(24.70)	(23.54)	(18.94)
Ln(Livesessions)	–0.080	–0.157	–0.102
	(–0.66)	(–1.39)	(–0.89)
Ln(Duration)		0.346**	–0.162
		(2.50)	(–0.74)
Ln(Products)		–0.624***	–0.833***
		(–3.48)	(–4.02)
LiveMode			–7.584**
			(–2.14)
Ln(Duration) × *LiveMode*			0.841**
			(2.49)
Ln(Products) × *LiveMode*			0.701*
			(1.73)
Constant	–6.514***	–5.983***	–1.734
	(–4.93)	(–3.24)	(–0.77)
Observations	365	365	365
R^2	0.749	0.787	0.793

Notes: t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Duration is significantly and positively correlated with *Sales*($p < 0.01$); *Products*is significantly and negatively correlated with *Sales* ($p < 0.01$). In the following paragraphs, we will conduct statistical tests to examine the impact of the duration of product introduction and the number of listed products on brand stores' sales performance.

5.2. Regression analysis

For the analysis in this section, we employ Stata 16.0 SE. The estimation results for the main effects of the duration of product introduction and the number of listed products on the sales performance of brand stores, along with the heterogeneity in these main effects across different live-streaming modes, are presented in Table 6.

As shown in Model 1 of Table 6, Ln*(Price)* is negatively correlated with Ln*(Sales)* (Beta = –0.350, $p < 0.01$). Ln*(Viewers)* (Beta = 0.149, $p < 0.05$), Ln*(Likes)* (Beta = 0.160, $p < 0.01$), Ln*(Livefans)* (Beta = 0.632, $p < 0.01$), and Ln*(Brandproducts)* (Beta = 1.818, $p < 0.01$) have positive effects on Ln*(Sales)* separately. There is no significant correlation between Ln*(Livesessions)* and Ln*(Price)* (Beta = –0.080, $p > 0.1$).

As shown in Model 2 of Table 6, Ln*(Duration)* is positively associated with Ln*(Sales)* (Beta = 0.346, $p < 0.05$), whereas Ln*(Products)* is negatively associated with Ln*(Sales)* (Beta = –0.624, $p < 0.01$). The regression analysis indicates that a longer time for product introduction helps increase brand stores' sales; the greater the number of listed products, the more significant the decline in the brand stores' sales. Thus, H1 and H2 were supported.

As shown in Model 3 of Table 6, the impact of the duration of product introduction and the number of listed products on sales performance differs with the live-streaming mode. Compared with Internet celebrity live streaming, the positive effect of the duration of product introduction on brand online stores' sales performance is stronger in brand shop live-streaming mode (Beta = 0.841, $p < 0.05$). Compared with brand shop live streaming, the negative effect of the number of listed products on brand online stores' sales performance is stronger in Internet celebrity live-streaming mode (Beta = 0.701, $p < 0.1$). As a result, H3 and H4 were supported.

In order to have a clearer understanding of the moderating effect, Fig. 2 and Fig. 3 were plotted based on the regression results of Model 3.

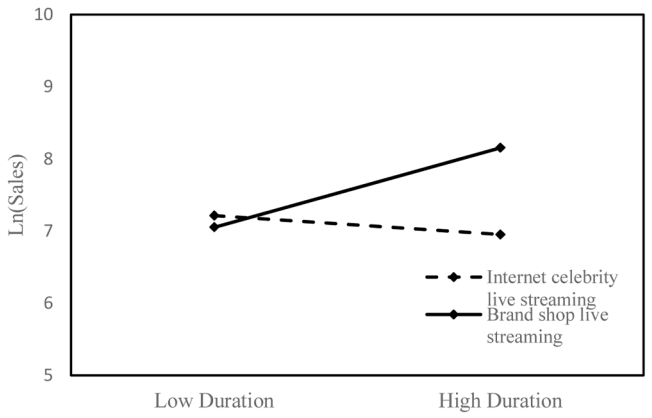


Fig. 2. Moderating effect of the duration of product introduction.

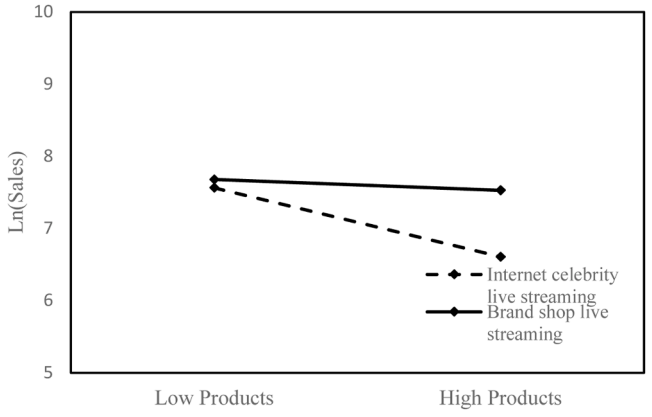


Fig. 3. Moderating effect of the number of listed products.

As shown in Fig. 2, in the brand shop live-streaming situation (illustrated by the solid line, slope = 0.679), the duration of product introduction exerts a more significant influence on sales volume than that in the Internet celebrity situation (dashed line, slope = –0.162), indicating that live-streaming mode moderated the relationship between the duration of product introduction and the sales volume in the way predicted by H3. Fig. 3 demonstrates that the slope was steeper when the live-streaming mode was Internet celebrity live streaming (dashed line, slope = –0.833 < –0.132). This finding suggests that the live-streaming mode acts as a moderator for the relationship between the listed number of products and sales volume, consistent with H4.

5.3. Propensity score matching method

Considering that the characteristics of different brand stores may affect our regression results, we employ the propensity score matching (PSM) method to eliminate differences between brand stores that choose brand shop live streaming and stores that choose Internet celebrity live streaming [60]. We utilize characteristics of the brands, such as *Storefans, Star, Freq, Capital*, and *Celebrities*, to perform logit regression on *Livemode* (the definitions of the variables and their descriptive statistics are presented in Appendix A). Subsequently, we employed *K*-nearest-neighbor matching, radius matching, and kernel matching within the PSM framework to pair brand stores that choose brand shop live streaming with those that choose Internet celebrity live streaming (the detailed matching results are presented in Tables A.2–A.4 and Figs. A.1–A.3 of the Appendix). After matching, the *p*-values for all variates exceeded 0.1, indicating that the matching balance was satisfied. The absolute value of the standardized differences for all variables was below 10 %, indicating that the matching results were valid [61]. Following the matching procedure, the brand stores that employ brand shop live streaming have characteristics similar to those of stores that employ Internet celebrity live streaming. In the final sample, we obtain a total of 160, 309, and 338 matched samples for *K*-nearest-neighbor matching, radius matching, and kernel matching methods, respectively.

Table 7 presents the results of the regression models utilizing

Table 7
Regression analysis (PSM).

	K-nearest neighbor matching (*N* = 1:6)			Radius matching (Radius = 0.04)			Kernel matching (Bandwidth = 0.06)		
Variables	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Ln(Price)	–0.374***	–0.672***	–0.693***	–0.422***	–0.574***	–0.613***	–0.377***	–0.535***	–0.567***
	(–2.98)	(–5.47)	(–5.68)	(–4.95)	(–7.04)	(–7.47)	(–4.81)	(–7.11)	(–7.51)
Ln(Viewers)	0.097	0.330***	0.317***	0.147**	0.238***	0.277***	0.176***	0.230***	0.263***
	(0.99)	(3.41)	(3.14)	(2.11)	(3.55)	(3.89)	(2.72)	(3.73)	(4.01)
Ln(Likes)	0.212**	0.212**	0.239***	0.183***	0.200***	0.225***	0.152**	0.189***	0.210***
	(2.20)	(2.46)	(2.76)	(2.71)	(3.20)	(3.61)	(2.42)	(3.24)	(3.60)
Ln(Livefans)	0.574***	0.479***	0.442***	0.658***	0.688***	0.595***	0.636***	0.687***	0.600***
	(4.25)	(3.92)	(3.28)	(6.95)	(7.89)	(6.02)	(7.00)	(8.21)	(6.37)
Ln(Brandproducts)	1.697***	1.672***	1.672***	1.869***	1.714***	1.677***	1.865***	1.684***	1.638***
	(15.24)	(16.73)	(14.25)	(21.60)	(20.85)	(16.98)	(23.11)	(21.76)	(17.45)
Ln(Livesessions)	–0.215	–0.206	–0.224	–0.151	–0.237*	–0.174	–0.147	–0.233**	–0.167
	(–1.13)	(–1.21)	(–1.31)	(–1.16)	(–1.96)	(–1.42)	(–1.18)	(–2.02)	(–1.43)
Ln(Duration)		0.549**	–0.445		0.388**	–0.262		0.365**	–0.180
		(2.52)	(–1.01)		(2.50)	(–1.06)		(2.53)	(–0.82)
Ln(Products)		–0.628**	–1.143***		–0.682***	–0.946***		–0.683***	–0.880***
		(–2.18)	(–3.00)		(–3.37)	(–4.08)		(–3.67)	(–4.19)
LiveMode			–12.867**			–9.714**			–8.787**
			(–2.43)			(–2.47)			(–2.37)
Ln(Duration) × LiveMode			1.389**			1.110***			0.994***
			(2.59)			(2.98)			(2.86)
Ln(Products) × LiveMode			1.129*			0.804*			0.742*
			(1.90)			(1.76)			(1.71)
Constant	–4.181**	–5.039	3.445	–6.434***	–6.207***	–1.042	–6.351***	–5.973***	–1.689
	(–2.24)	(–1.63)	(0.80)	(–4.50)	(–3.00)	(–0.41)	(–4.58)	(–3.05)	(–0.73)
Observations	160	160	160	309	309	309	338	338	338
R^2	0.701	0.763	0.775	0.723	0.770	0.778	0.728	0.774	0.782

Notes: t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

matched samples. As indicated in Models 5, 8, and 11 of Table 7, the duration of product introduction remains positively associated, whereas the number of listed products remains negatively associated, both statistically significant at the 5 % and 1 % levels, respectively. Furthermore, as demonstrated in Models 6, 9, and 12 of Table 7, the effect of the duration of product introduction and the number of listed products on sales performance differs across the live-streaming mode. The findings suggest that the differences in characteristics between the brand stores that employ brand shop live streaming and those that employ Internet celebrity live streaming do not drive our main results. In summary, after addressing the selection bias inherent in stores' self-selection of live-streaming modes, we confirmed the robustness of our results.

5.4. Robustness checks

We changed the dependent variable of Models 1, 2, and 3 from sales volume to sales revenue. Those robustness test results are shown in Table 8. In alignment with the conclusions obtained in Table 6, the duration of product introduction exhibits a positive effect on brand online stores' sales performance (Beta = 0.412, $p < 0.01$), while the number of listed products demonstrates a negative effect on brand online stores' sales performance (Beta = –0.609, $p < 0.01$). Moreover, compared with Internet celebrity live streaming, the positive effect of the duration of product introduction on brand online stores' sales performance is enhanced in brand shop live-streaming mode (Beta = 0.906, $p < 0.05$). No significant differences in the effect of the number of listed products on sales performance are observed across the live-streaming mode (Beta = 0.701, $p > 0.1$).

To mitigate the potential bias arising from unobserved brand-level factors, we incorporated several relevant brand characteristics as control variables in our analysis. These include *BrandSize* (classified into large brands and small brands, where brands with >2500,000 followers on their store pages are considered large brands, represented by 1), *Years* (measured by the number of years since the brand was established), and *Capital* (registered capital of the brand). These variables capture key aspects of a brand's resources and market presence, which could be correlated with our outcome variable (brand sales). After these control variables are incorporated, our results remain substantively unchanged (as shown in Table 9), suggesting that our findings are robust to the inclusion of these observable brand-level characteristics.

Table 8
Extended analysis using sales revenue as dependent variable.

Variables	Model 13	Model 14	Model 15
Ln(Price)	0.536***	0.357***	0.324***
	(6.81)	(4.64)	(4.17)
Ln(Viewers)	0.166**	0.237***	0.266***
	(2.50)	(3.67)	(3.84)
Ln(Likes)	0.148**	0.180***	0.200***
	(2.25)	(2.93)	(3.25)
Ln(Livefans)	0.727***	0.750***	0.664***
	(7.93)	(8.77)	(6.77)
Ln(Brandproducts)	1.814***	1.657***	1.608***
	(23.14)	(21.89)	(17.72)
Ln(Livesessions)	–0.175	–0.261**	–0.201
	(–1.36)	(–2.18)	(–1.64)
Ln(Duration)		0.412***	–0.115
		(2.80)	(–0.49)
Ln(Products)		–0.609***	–0.819***
		(–3.19)	(–3.70)
LiveMode			–8.064**
			(–2.14)
Ln(Duration) × LiveMode			0.906**
			(2.51)
Ln(Products) × LiveMode			0.701
			(1.62)
Constant	–6.981***	–6.886***	–2.563
	(–4.96)	(–3.50)	(–1.06)
Observations	365	365	365
R^2	0.778	0.811	0.816

Notes: *t*-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9
Extended analysis by adding brand characteristics as control variables.

Variables	Model 16	Model 17
Ln(Price)	–0.530***	–0.555***
	(–7.25)	(–7.51)
Ln(Viewers)	0.209***	0.235***
	(3.34)	(3.55)
Ln(Likes)	0.192***	0.216***
	(3.31)	(3.71)
Ln(Livefans)	0.644***	0.569***
	(6.88)	(5.67)
Ln(Brandproducts)	1.667***	1.612***
	(22.47)	(18.80)
Ln(Livesessions)	–0.155	–0.094
	(–1.35)	(–0.81)
Ln(Duration)	0.322**	–0.179
	(2.25)	(–0.80)
Ln(Products)	–0.618***	–0.829***
	(–3.44)	(–3.98)
BrandSize	0.023	–0.061
	(0.11)	(–0.29)
Ln(Years)	0.130	0.123
	(1.17)	(1.12)
Ln(Capital)	–0.022	–0.018
	(–0.61)	(–0.50)
LiveMode		–7.573**
		(–2.13)
Ln(Duration) × LiveMode		0.842**
		(2.47)
Ln(Products) × LiveMode		0.697*
		(1.72)
Constant	–5.735***	–1.785
	(–2.87)	(–0.75)
Observations	365	365
R^2	0.787	0.793

Notes: *t*-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6. Discussion and conclusions

6.1. Discussion

With the development of live-streaming e-commerce, numerous brands are exploring live streaming as an innovative sales channel. However, the real-time nature of live streaming, coupled with promotional content and a large volume of products, may overwhelm viewers and diminish benefits. From the dual perspectives of information overload theory and trust theory, this paper utilizes the sales data of beauty and skincare products during the "Double 11″ shopping festival of Taobao in the year 2020 and examines how the effect of the duration of product introduction and the number of listed products on sales performance differs across live-streaming modes. The findings are discussed below.

First, the duration of product introduction has a positive impact on brand online stores' sales performance. Owing to the untouchable nature of live-streaming e-commerce, the streamer's introduction of the product is a crucial way for viewers to gather information about it. Previous studies have demonstrated that the provision of a certain amount of product information can help consumers in making purchase decisions [62]. Specifically, the professional presentation and explanation of products via streaming enables consumers to handle product information easily, thereby reducing their overload levels and influencing their purchasing decisions positively [54].

Second, the number of listed products has a negative impact on brand online stores' sales performance. The more choices consumers have, the more information they need to process, which forces them to invest more time and energy. When the cognitive cost to consumers increases, it imposes a psychological burden on them, reducing their

willingness to make purchases [55–57].

Third, the live-streaming mode moderates the relationship between the duration of product introduction and brand online stores' sales performance. That is, compared with that for Internet celebrity live streaming, the positive effect of the duration of product introduction on brand online stores' sales performance is stronger in the brand shop live-streaming mode. The duration of product introduction serves as a critical factor enabling consumers to obtain adequate information, thus reducing perceived purchase risks [52]. However, the effectiveness of this factor varies significantly with the level of trust. Within the context of Internet celebrity live streaming characterized by lower trust, extending the product presentation time often fails to alleviate consumers' concerns fully regarding the brand or product quality. Consequently, the positive effect of the duration of introduction on sales becomes less pronounced.

Fourth, the live-streaming mode moderates the relationship between the number of listed products and brand online stores' sales performance. That is, compared with that for brand shop live streaming, the negative effect of the number of listed products on sales performance is stronger in Internet celebrity live-streaming mode. When consumers hold high levels of trust in a product they are considering, the adverse effects of information overload on their purchase decision are mitigated. This is analogous to a decision shortcut, where, relying on established trust, consumers are exempt from the need for extensive information processing regarding redundant or overwhelming data.

6.2. Theoretical implications

These findings offer some important theoretical implications. First, while prior research has established information overload as a key challenge in e-commerce, this study advances the theory by demonstrating its contingent and moderated effects within the dynamic, real-time context of live streaming. We move beyond a simple application of the theory by investigating how specific information characteristics—namely, the duration of product introduction and the number of listed products—impact consumer cognitive load. Our findings reveal that these factors do not affect all consumers uniformly. Instead, their influence is significantly moderated by the live-streaming mode. This contextualization extends the applicability of information overload theory, demonstrating that its explanatory power is not universal but is shaped by the unique interactive and synchronous nature of the live-streaming environment and the underlying trust mechanisms it activates.

Second, this paper distinguishes two live-streaming e-commerce modes and explores the trust mechanisms of consumers for these modes. Acknowledging that the literature on trust has proposed various typologies [21,63], we contend that the unique dynamics of the live-streaming e-commerce context calls for a new framework. Therefore, we introduce and empirically validate a novel framework that differentiates between trust in Internet celebrities and trust in brand shops, as detailed in Table 3. While trust in Internet celebrities is based on the perceived attractiveness, trustworthiness, and professionalism of the individual KOL (often an indirect form of trust, where trust in the person, who might be an actor, celebrity, or influential blogger, transfers to the product they endorse), trust in Internet brand shops is rooted in consumers' perceptions of a brand's product quality, service capabilities, and overall reliability (a more direct form of trust based on the brand's attributes). This distinction significantly extends the theory of online trust, moving its application from "search" and "browsing" based e-commerce to the more interactive and synchronous selling format of live-streaming e-commerce.

Third, we integrate information overload theory and trust theory to construct a holistic predictive framework for live-streaming consumer behavior. We propose that the relationship between information processing demands (e.g., product duration, number of products) and consumer outcomes is not direct but is critically mediated by the specific trust mechanism activated by the live-streaming mode. This integrative lens clarifies boundary conditions for information overload—when it harms consumer decision-making and when the harm is mitigated. For instance, consumer trust in a brand can serve as a stronger buffer against information overload resulting from product proliferation than trust in an Internet celebrity. This insight addresses a critical gap in the extant literature by explicating the mechanisms (why) and contingencies (when) shaping information environment effectiveness in live-streaming, which prior studies overlooked by examining overload or trust in isolation. Consequently, this study provides a more nuanced theoretical understanding of consumer behavior in live-streaming e-commerce, offering valuable insights for the development of context-specific operational and managerial strategies for both brand managers and influencers.

6.3. Practical implications

The practical implications of this study are delineated below. First, it is recommended that brand stores actively employ the brand shop live-streaming mode to enhance service quality for consumers. They should provide training sessions to ensure that branded streamers possess sufficient brand knowledge, and, if necessary, evaluate their professionalism following the training. Furthermore, brand streamers ought to place particular emphasis on elucidating key product information in order to facilitate consumers' rapid comprehension of products, thereby enabling them to make well-informed purchase decisions. Second, Internet celebrity live streaming is also an important way for brand stores to obtain live-streaming revenue. In seeking collaboration with Internet celebrities, beyond merely considering the follower base of the Internet celebrity, brands should also limit the number of products listed in the Internet celebrity's live-streaming sessions. An excessive number of listed products can impair the brand's sales performance severely.

6.4. Limitations and future research

This study has some limitations. First of all, this paper utilized the sales data of beauty and skincare products during the "Double 11″ shopping festival of Taobao in the year 2020. In the future, the robustness of our findings can be triangulated using data from other online platforms and/or product categories. In addition, the characteristics of the streamers that may affect the marketing effects of live streaming should be controlled. Information such as gender, age, professionalism, entertainment, and trustworthiness of the streamers can be collected using a questionnaire method in future studies to examine their impacts on sales performance. Furthermore, to better account for and mitigate the influence of unobservable factors, future research could employ panel data or natural experiment designs.

Author statement

This paper (or closely related research) has not been published or accepted for publication.

CRediT authorship contribution statement

Jianping Zheng: Writing – original draft, Visualization, Investigation, Formal analysis. **Youwei Wang:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **Yifan Dou:** Writing – review & editing, Supervision, Data curation. **Hong Ling:** Writing – review & editing, Supervision.

Declaration of competing interest

The nominees for senior editors, associate editors, and reviewers have no conflict of interest with any of the authors of the paper being submitted.

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Supplementary materials

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References

[1] Y.S. Wang, User experiences in live video streaming: a netnography analysis, Internet Res. 29 (4) (2019) 638–656.
[2] B. Lu, Z. Chen, Live streaming commerce and consumers' purchase intention: an uncertainty reduction perspective, Inf. Manag. 58 (7) (2021) 103509.
[3] China Commercial Industry Research Institute, China's live-streaming e-commerce industry: market analysis and investment risk trend forecast research report (2025–2030) (accessed on, https://www.askci.com/news/chanye/20250625/095656275081661551433228.shtml, August 12, 2025 (accessed on.
[4] D. Wang, X.R. Luo, Y. Hua, J. Benitez, Big arena, small potatoes: a mixed-methods investigation of atmospheric cues in live-streaming e-commerce, Decis. Support Syst. 158 (2022) 113801.
[5] Z. Lu, H. Xia, S. Heo, D. Wigdor, You watch, you give, and you engage: a study of live streaming practices in China, in: Conf. Hum. Factors Comput. Syst. - Proc, 2018, pp. 1–13.
[6] A. Wongkitrungrueng, N. Assarut, The role of live streaming in building consumer trust and engagement with social commerce sellers, J. Bus. Res. 117 (2020) 543–556.
[7] 36 Kr, annual research report on chinese online live streaming e-commerce industry in 2020. https://36kr.com/p/986332005917833, 2020 (accessed on June 5, 2023).
[8] W. Wang, M. Huang, S. Zheng, L. Lin, L. Wang, The impact of broadcasters on consumer's intention to follow livestream brand community, Front. Psychol. 12 (2022) 810883.
[9] H. Chen, Y. Dou, Y. Xiao, Understanding the role of live streamers in live-streaming e-commerce, Electron. Commer. Res. Appl. 59 (2023) 101266.
[10] X. Yang, Y. Liu, J. Dong, S. Li, Impact of streamers' characteristics on sales performance of search and experience products: evidence from Douyin, J. Retail. Consum. V. 70 (2023) 103155.
[11] A.S. Krishen, R.L. Raschke, P. Kachroo, A feedback control approach to maintain consumer information load in online shopping environments, Inf. Manag. 48 (8) (2011) 344–352.
[12] N.K. Malhotra, Information load and consumer decision making, J. Consum. Res. 8 (4) (1982) 419–430.
[13] J. Jacoby, D.E. Speller, C.K. Berning, Brand choice behavior as a function of information load: replication and extension, J. Consum. Res. 1 (1) (1974) 33–42.
[14] X. Gao, X.Y. Xu, S.M.U. Tayyab, Q. Li, How the live streaming commerce viewers process the persuasive message: an ELM perspective and the moderating effect of mindfulness, Electron. Commer. Res. Appl. 49 (2021) 101087.
[15] R. Sethuraman, J.C. Gázquez-Abad, F.J. Martínez-López, The effect of retail assortment size on perceptions, choice, and sales: review and research directions, J. Retail. 98 (1) (2022) 24–45.
[16] D. Kuksov, M. J, Villas-Boas, when more alternatives lead to less choice, Mark. Sci. 29 (3) (2010) 507–524.
[17] C. Lou, S.S. Tan, X. Chen, Investigating consumer engagement with influencer-vs. brand-promoted ads: the roles of source and disclosure, J. Interact. Advert. 19 (3) (2019) 169–186.
[18] C. Chen, Y. Hu, Y. Lu, Y. Hong, Everyone can be a star: quantifying grassroots online sellers' live streaming effects on product sales, in: Proc. 52nd Hawaii Intl. Conf. Sys. Sci, 2019, pp. 4493–4498.
[19] S. Yuan, L. Liu, B. Su, H. Zhang, Determining the antecedents of mobile payment loyalty: cognitive and affective perspectives, Electron. Commer. Res. Appl. 41 (2020) 100971.
[20] A. Chaudhuri, M.B. Holbrook, The chain of effects from brand trust and brand affect to brand performance: the role of brand loyalty, J. Mark. 65 (2) (2001) 81–93.
[21] S. Ba, P.A. Pavlou, Evidence of the effect of trust building technology in electronic markets: price premiums and buyer behavior, MIS Quart. 26 (3) (2002) 243–268.
[22] P. Xu, B. Cui, B. Lyu, Influence of streamer's social capital on purchase intention in live streaming E-commerce, Front. Psychol. 12 (2022) 748172.
[23] Y. Sun, X. Shao, X. Li, Y. Guo, K. Nie, How live streaming influences purchase intentions in social commerce: an IT affordance perspective, Electron. Commer. Res. Appl. 37 (2019) 100886.
[24] J. Cai, D.Y. Wohn, A. Mittal, D. Sureshbabu, Utilitarian and hedonic motivations for live streaming shopping, in: 2018 ACM Int. Conf. on Interact. Exp. for TV and online video, 2018, pp. 81–88.
[25] H. Chen, H. Chen, X. Tian, The dual-process model of product information and habit in influencing consumers' purchase intention: the role of live streaming features, Electron. Commer. Res. Appl. 53 (2022) 101150.
[26] H.J. Park, L.M. Lin, The effects of match-ups on the consumer attitudes toward internet celebrities and their live streaming contents in the context of product endorsement, J. Retail. Consum. v 52 (2020) 101934.
[27] L. Meng, F.J. Liu, S.Y. Chen, K. Duan, Can I evoke you? A study on the influence mechanism of information source characteristics of different types of live broadcasting celebrity on consumers' willingness to purchase, Nankai Bus. Rev. 23 (01) (2020) 131–143 (In Chinese).
[28] M.X. Huang, Y.Q. Ye, W. Wang, The interaction effect of broadcaster and product type on consumers' purchase intention and behaviors in livestreaming shopping, Nankai Bus. Rev. 26 (02) (2023) 188–198 (In Chinese).
[29] D. Liu, J. Yu, Impact of perceived diagnosticity on live streams and consumer purchase intention: streamer type, product type, and brand awareness as moderators, Inf. Technol. Manag. 25 (3) (2024) 219–232.
[30] C. Zhang, S. Pan, Y. Zhao, More is not always better: examining the drivers of livestream sales from an information overload perspective, J. Res. Consum. v 77 (2024) 103651.
[31] D. Wang, X.R. Luo, Y. Hua, J. Benitez, Customers' help-seeking propensity and decisions in brands' self-built live streaming E-commerce: a mixed-methods and fsQCA investigation from a dual-process perspective, J. Bus. Res. 156 (2023) 113540.
[32] M.J. Eppler, J. Mengis, The concept of information overload—A review of literature from organization science, accounting, marketing, MIS, and related disciplines, Inf. Soc. 20 (5) (2004) 325–344.
[33] D. Bawden, L. Robinson, Information overload: an introduction, Oxford research encyclopedia of politics, Oxford University Press, June 30, 2020.
[34] R.S. Wurman, Information anxiety 2, Que Pub., 2001.
[35] S. Fu, H. Li, Y. Liu, H. Pirkkalainen, M. Salo, Social media overload, exhaustion, and use discontinuance: examining the effects of information overload, system feature overload, and social overload, Inf. Process. Manag. 57 (6) (2020) 102307.
[36] R.L. Meier, Communications overload: proposals from the study of a university library, Admin. Sci. Quart. 7 (4) (1963) 521–544.
[37] T.E. Vollmann, Cutting the gordian knot of misguided performance measurement, Ind Manage. Data. Syst. 91 (1) (1991) 24–26.
[38] C. Klausegger, R.R. Sinkovics, H. Zou, Information overload: a cross-national investigation of influence factors and effects, Mark. Intell. Plan 25 (7) (2007) 691–718.
[39] B. Scheibehenne, R. Greifeneder, P.M. Todd, What moderates the too-much-choice effect? Psychol. Mark. 26 (3) (2009) 229–253.
[40] N.H. Lurie, Decision making in information-rich environments: the role of information structure, J. Consum. Res. 30 (4) (2004) 473–486.
[41] D. Gefen, E. Karahanna, D.W. Straub, Trust and TAM in online shopping: an integrated model, MIS Quart 27 (1) (2003) 51–90.
[42] C. Liu, Z. Bao, C. Zheng, Exploring consumers' purchase intention in social commerce: an empirical study based on trust, argument quality, and social presence, Asia Pac. J. Mark. Logist. 31 (2) (2019) 378–397.
[43] S.M. Lee, S.J. Lee, Consumers' initial trust toward second-hand products in the electronic market, J. Comput. Inf. Syst. 46 (2) (2005) 85–98.
[44] M. Tajvidi, Y. Wang, N. Hajli, P.E. Love, Brand value Co-creation in social commerce: the role of interactivity, social support, and relationship quality, Comput. Hum. Behav. 115 (2021) 105238.
[45] Y.H. Chen, M.C. Chen, C.J. Keng, Measuring online live streaming of perceived servicescape: scale development and validation on behavior outcome, Internet Res. 30 (3) (2020) 737–762.
[46] K.H. Lim, C.L. Sia, M.K.O. Lee, I. Benbasat, Do I trust you online, and if so, will I buy? An empirical study of two trust-building strategies, J. Manage. Inf. Syst. 23 (2) (2006) 233–266.
[47] M. Zhang, Y. Liu, Y. Wang, L. Zhao, How to retain customers: understanding the role of trust in live streaming commerce with a socio-technical perspective, Comput. Hum. Behav. 127 (2022) 107052.
[48] R. Chinomona, E.T. Maziriri, The influence of brand awareness, brand association and product quality on brand loyalty and repurchase intention: a case of male consumers for cosmetic brands in South Africa, J. Bus. Retail Manag. Res. 12 (1) (2017) 143–154.
[49] E. Delgado-Ballester, J. L.Munuera-Alemán, brand trust in the context of consumer loyalty, Eur. J. Mark. 35 (11/12) (2001) 1238–1258.
[50] S.A. Ruly, M.N.P. Riski, M.I.A.C. Saisaria, The role of brand trust in mediating the correlation between the use of celebrity endorsers and online consumer purchase intentions, Russ. J. Agric. Socio-Econ. Sci. 115 (7) (2021) 131–137.
[51] R. Husain, J. Paul, B. Koles, The role of brand experience, brand resonance and brand trust in luxury consumption, J. Retail. Consum. v 66 (2022) 102895.
[52] S. Teng, K. Wei Khong, W. Wei Goh, A. Yee Loong Chong, Examining the antecedents of persuasive eWOM messages in social media, Online Inf. Rev. 38 (6) (2014) 746–768.
[53] W. Afzal, D. Roland, M.N. Al-Squri, Information asymmetry and product valuation: an exploratory study, J. Inf. Sci. 35 (2) (2009) 192–203.
[54] H.F. Hu, A.S. Krishen, When is enough, enough? Investigating product reviews and information overload from a consumer empowerment perspective, J. Bus. Res. 100 (2019) 27–37.
[55] J.T. Gourville, D. Soman, Overchoice and assortment type: when and why variety backfires, Mark. Sci. 24 (3) (2005) 382–395.
[56] J.M. Adriatico, A. Cruz, R.C. Tiong, C.R. Racho-Sabugo, An analysis on the impact of choice overload to consumer decision paralysis, J. Econ. Fin. Acc. Stud. 4 (1) (2022) 55–75.
[57] J. Beneke, Are consumers really bewildered by overchoice? An experimental approach to the tyranny of "too much, J. Food Prod. Mark. 21 (1) (2015) 90–101.

[58] B. Schwartz, The paradox of choice, in: S. Joseph (Ed.), Positive psychology in practice: promoting human flourishing in work, health, education, and Everyday Life, John Wiley & Sons, Hoboken, NJ, 2015, pp. 121–138.
[59] Ju Meli, Tmall's most complete 2020 report card: sets exceed 30 billion, masks, essences, and facial makeup exceed 20 billion. https://www.36kr.com/p/1075275277153410, 2021 (accessed on June 5, 2023).
[60] Y. Cai, H. Lyu, M. Peng, The role of sponsor representatives in SEO underpricing: evidence from China, Int. Rev. Econ. Financ. 89 (2024) 30–45.
[61] P.R. Rosenbaum, D.B. Rubin, Constructing a control group using multivariate matched sampling methods that incorporate the propensity score, Amer. Stat. 39 (1) (1985) 33–38.
[62] C.P. Furner, R. Zinko, Z. Zhu, Electronic word-of-mouth and information overload in an experiential service industry, J. v. Theory Pr. 26 (6) (2016) 788–810.
[63] B.L. Connelly, T.R. Crook, J.G. Combs, D.J. Ketchen Jr, H. Aguinis, Competence- and integrity-based trust in interorganizational relationships: which matters more? J. Manage 44 (3) (2018) 919–945.

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